Exponential & Logarithmic Functions Maze **ANSWER KEY**

Name _____ Period _____

Start in the top-left room. Evaluate or solve each problem (round only when told to, and only at the end). Go through the door that shows your answer. Show your work on the answer grid.

START

1 Evaluate.

$\log_3 81 - \log_2 32$

→ **−1** → **2** Expand. Assume $x > 0$.

$\log_5(25x^2)$

→ **2 + 2log₅x** → **3** Solve for x.

$9^x = 27^{x-1}$

→ **3** → **4** Solve. Round to the nearest hundredth.

$5^x = 40$

↓ 49

Evaluate.

$\log_9 27$

↓ 18

↓ 33 (ln 6)

Solve for x.

$3 \cdot 2^{x-1} = 96$

↓ 5 or −2

7 Solve. Check for extraneous solutions.

$\log x + \log(x - 3) = 1$

→ **5**

↓ $2 + \log_5 x$

↓ $2\log_5 x$

Write as a single logarithm.

$3 \ln 2 + \ln 5 - \ln 10$

↓ $2,700

6 $2,000 is invested at 5% compounded quarterly for 6 years.

Find the balance to the nearest dollar.

→ $2,680

↓ 10

8 Money grows at 4% compounded continuously.

How many years to double? Nearest tenth.

→ 1

↓ −3

↓ 0.458

5 Use change of base. Round to the nearest thousandth.

$\log_6 50$

↓ 12

↓ $\frac{3}{5}$

9 Solve. Check for extraneous solutions.

$\log_2(x - 5) = \log_2(2x + 1)$

→ **no solution**

↓ 7.5

↓ 1.699

Evaluate.

$\ln e^7 - \log 0.01$

↓ −6

↓ 11

Solve for x.

$\ln(2x) = 3$

↓ 20

10 A sample has a half-life of 12 days.

80 mg now. How many mg remain after 30 days? Nearest tenth.

→ **14.1**

↓ 0.44

↓ $\frac{3}{e^2}$

↓ e^3

FINISH

You made it!

2.29 · **2.183** · **$2,695** · **17.3**

Exp. & Log. Functions Maze: Answer Grid **ANSWER KEY**

Name _____ Period _____

Write each answer in the order you found it, and show your work for that room.

1	Answer: **-1**	$\log_3 81 = 4,\ \log_2 32 = 5 \rightarrow 4 - 5 \rightarrow -1$
2	Answer: **$2 + 2\log_5 x$**	$\log_5 25 + \log_5 x^2 = 2 + 2\log_5 x \rightarrow 2 + 2\log_5 x$
3	Answer: **3**	$3^{2x} = 3^{3(x-1)} \rightarrow 2x = 3x - 3 \rightarrow 3$
4	Answer: **2.29**	$x = \frac{\ln 40}{\ln 5} \approx \frac{3.689}{1.609} \rightarrow 2.29$
5	Answer: **2.183**	$\log_6 50 = \frac{\ln 50}{\ln 6} = \frac{3.912}{1.792} \rightarrow 2.183$
6	Answer: **$2,695**	$A = 2000(1 + \frac{0.05}{4})^{4 \cdot 6} = 2000(1.0125)^{24} \rightarrow \$2{,}695$
7	Answer: **5**	$x(x - 3) = 10 \rightarrow (x - 5)(x + 2) = 0$; $x = -2$ makes $\log x$ undefined $\rightarrow 5$
8	Answer: **17.3**	$2P = Pe^{0.04t} \rightarrow \ln 2 = 0.04t \rightarrow t = \frac{\ln 2}{0.04} \rightarrow 17.3$
9	Answer: **no solution**	$x - 5 = 2x + 1 \rightarrow x = -6$, but then $x - 5 < 0 \rightarrow$ no solution
10	Answer: **14.1**	$80(\frac{1}{2})^{\frac{30}{12}} = 80(\frac{1}{2})^{2.5} \rightarrow 14.1$